

SURF facility status and DUNE response

A reference architecture for monitoring, orderly DAQ shutdown, detector protection, and local protection



Facility information provides context. Action authority stays with the system that owns the response.

Why the interface matters

Underground operations depend on facility conditions, while the experiment and the facility retain separate control authority.

OPERATIONAL CONTINUITY

Power and reserve

Loss of normal power can limit the time available for an orderly DAQ stop.

SOURCE AUTHORITY

Fire and air

SURF owns fire, low-oxygen, and ventilation alarms and the personnel response.

APPROVED RESPONSE

Cryogenics, cooling, and water

DUNE action follows an approved cause and effect owned by the source and detector systems.

OPERATING CONTEXT

Network, weather, and access

These conditions inform operations. A lost information path remains visible.

Four functions remain distinct

The same facility event can appear on several paths without giving those paths the same authority.

CONTEXT

Facility monitoring

Source → BMS → boundary → Slow Controls. The source owner retains alarm and reset authority.

APPROVED ACTION

Detector protection

Approved source → open interface → DPS. DPS owns permit, inhibit, trip, latch, and reset.

OPERATIONAL ACTION

Orderly DAQ shutdown

UPS state → backed automation → Run Control. DAQ owns run state; this path carries no protection credit.

AT THE LOAD

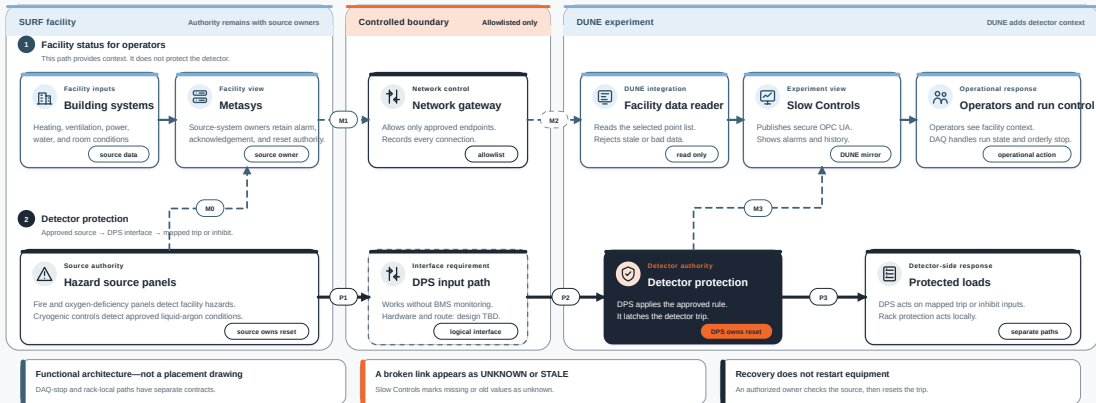
Local protection

Local sensor → local logic → affected load. The equipment owner retains action and restart authority.

How facility signals reach DUNE

Diagram 01 • Reference architecture

The building management system shares status. The Detector Protection System (DPS) handles approved trips.



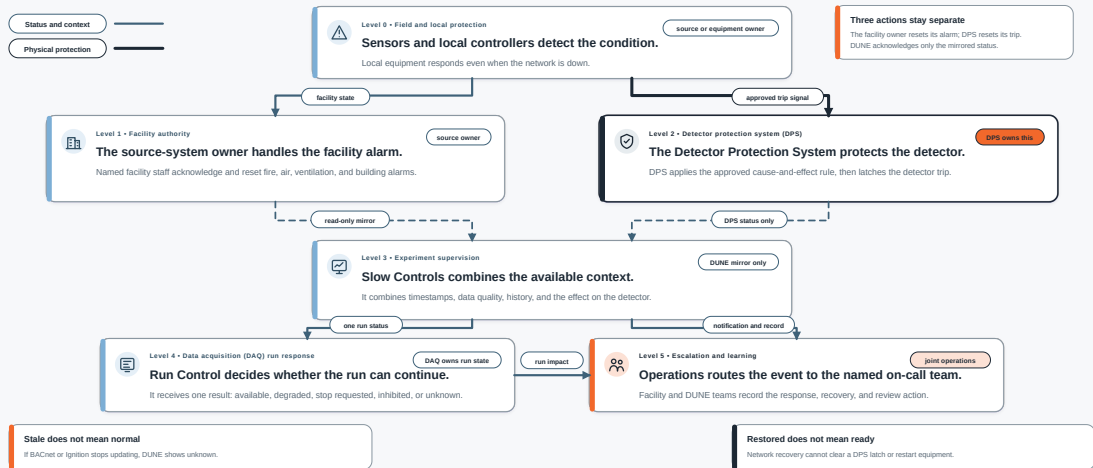
MONITORING M0 source status • M1 selected building-management points • M2 BACnet/IP boundary • M3 DPS status

PROTECTION P1 approved source state • P2 DPS interface—physical design TBD • P3 mapped detector trip or inhibit

Who owns the alarm, action, and reset

A status can move through the chain. The right to acknowledge or reset it stays with the system that owns the action.

Diagram 02 • Alarm authority



What crosses the boundary

DUNE receives facility state, not authority over the source alarm or reset.

FACILITY STATUS

BACnet/IP

Selected source points carry value, state, source time, quality, and freshness.

DUNE VIEW

OPC UA

Stable tags add units, age, detector context, alarms, and history.

SERVICE HEALTH

SNMPv3

Telemetry exposes network health and the health of backed services.

EVENT CHRONOLOGY

NTP and protected logs

Source time and receive time support a coherent account of an event.

What changes when the facility changes

Diagram 03 • Failure response

For each condition, the map shows ownership, operator context, DUNE response, and validation evidence.

FACILITY CONDITION	WHAT OPERATORS SEE	WHAT DUNE DOES	VALIDATION EVIDENCE
 Power and reserve ELECTRICAL • SURF	Utility, transfer switch, generator, backup-supply state, and remaining reserve	An orderly data-acquisition stop is requested. This operational path is not a detector trip.	EVIDENCE Transfer test Loaded UPS runtime
 Network or BMS link NETWORKING • CONTROLS	The failed link and affected points; values change to UNKNOWN or STALE	Detector and local protection remain available. State facility data is never treated as current.	EVIDENCE Per-link failure injection Recovery-behavior verification
 Fire, low oxygen, or ventilation FACILITY SAFETY • SURF	Source alarm, trouble, bypass, area, acknowledgement, and time	Facility teams handle people and facility response. DPS applies only an approved mapped action.	EVIDENCE Zone-by-zone test Trouble and bypass
 Cryogenic or liquid-argon state CRYOGENICS • DETECTOR SAFETY	Named level, pressure, relief, boiling-risk, and oxygen-deficiency states	The response follows the approved HV or DPS mapping. No generic liquid-argon leak signal is assumed.	EVIDENCE Source definition Cause-and-effect test
 Cooling or water FACILITIES • EQUIPMENT OWNER	Area, affected equipment, water or cooling state, and owner	Rack or equipment-local protection acts first. DPS acts only where the matrix requires it.	EVIDENCE AHU, water, and sump Local-sensor test
 Weather or access restriction SURF OPERATIONS	Defined access, shaft, power, cooling, or communications restriction	Raw weather provides context only. Action follows a defined SURF operational state.	EVIDENCE Tabletop scenario Operational-state injection
Orderly stop • data acquisition and automation Uses tested UPS reserve for a controlled shutdown. Detector trip • Detector Protection System Applies approved cause and effect; latches where required. Local protection • rack or equipment Acts close to the threatened load and reports status upward.			

Unavailable information stays visible

A failed data path changes what DUNE knows; it does not change the source condition or authorize a restart.

GOOD

Current

The value is current and carries valid source, time, and quality information.

STALE

Aging

The value remains historical context while its age and alarm stay visible.

UNKNOWN

Absent

The source, link, or service cannot provide a trustworthy current state.

RECOVERY

Restored

Updates resume while latches, inspection, and restart authority stay with their owners.

What is established—and what remains open

The functional boundary is clear; its remaining interface facts belong with named owners.

ESTABLISHED STARTING POINT

SURF retains source-alarm authority.

The BMS-to-DUNE path is monitoring only.

DAQ, DPS, and local action remain separate.

The diagrams define functions, not equipment.

OPEN ARCHITECTURE QUESTION

Authoritative points and state definitions.

Live service boundary, security behavior, and ownership.

Conditions for an orderly stop or detector action.

Interface medium, supervision, and failure behavior.

Architecture acceptance

Acceptance connects four properties: authority, information, failure behavior, and verification.

ACCEPTANCE DIMENSION

Authority

One owner is identifiable for the source alarm, run state, detector action, and local action.

ACCEPTANCE DIMENSION

Information

Source, time, quality, age, and state meaning remain intact from origin to operator view.

ACCEPTANCE DIMENSION

Failure behavior

Lost or stale data is visible; monitoring loss does not remove detector or local protection.

ACCEPTANCE DIMENSION

Verification

Test evidence covers the boundary, selectivity, loss, and recovery behavior.